Lecture 6A - Quasi-Experimental Designs

difference-in-differencess

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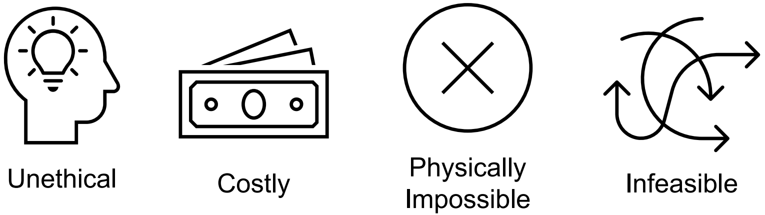
Outline

1. Quasi-experimental design
2. The canonical DID
    - The intuition behind DID
    - The canonical DID
    - Understanding parallel trend assumption
3. Estimating ATT
4. Dealing with the parallel trend assumption
5. Recent developments in DID

1. Quasi-experimental design
2. The canonical DID
3. Estimating ATT
4. Dealing with the parallel trend assumption
5. Recent developments in DID

Randomized experiments are powerful, but...

In many scenarios, we cannot do randomized experiments.



Quasi-experiments

Definition (Quasi-experiments)

Research designs that resemble experimental designs but lack full random assignment of participants to groups.

Quasi-experiments

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Research designs that resemble experimental designs but lack full random assignment of participants to groups.

Different designs of quasi-experiments

- Interrupted time-series design
- Event studies
- Case-control design
- **difference-in-differences (DID)**
- **Regression discontinuity design (RDD)**
- ...

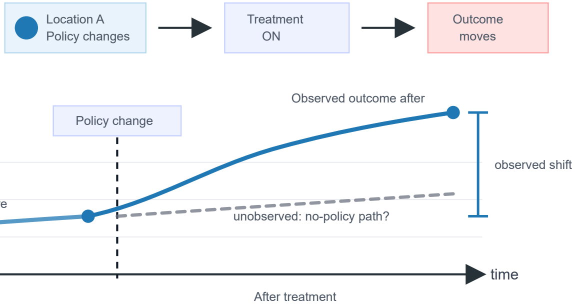
DID as an experimental design

Full randomization

Often costly, disruptive, or hard to implement.

A minimal field intervention

Change the treatment or policy in one location or group.
Observe outcomes before and after the change.

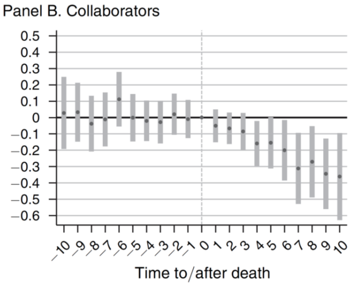


When can we learn the treatment effect of location or group A?

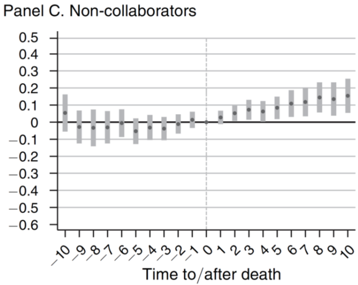
Examples of quasi-experiments...

Azoulay et al. (2019): How the **premature death** of eminent life scientists alters the vitality of their fields.

- The flow articles by **collaborators** into affected fields **decrease**.
- The flow of articles by **non-collaborators increases** markedly.



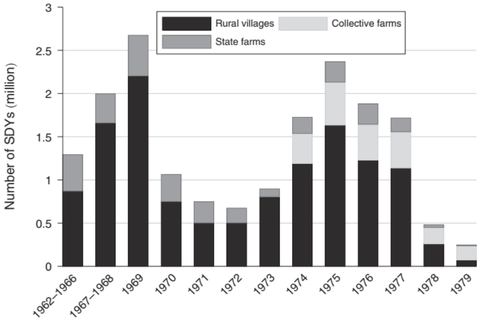
(a) Collaborators



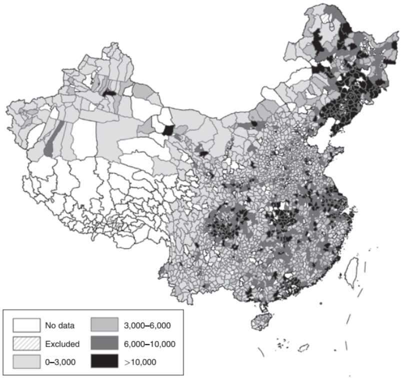
(b) Non-collaborators

The send-down movement in China

Chen et al. (2020) studies how the **send-down movement** in China contributes to education achievement in rural regions.



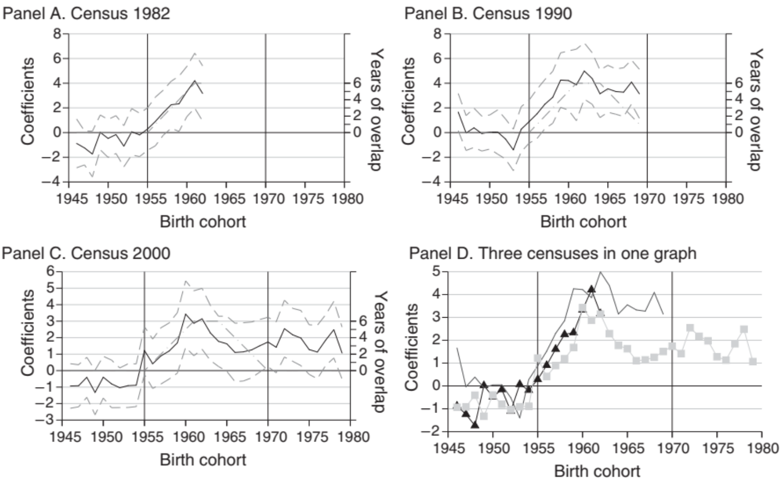
(a) No. of Sent-down Youths



(b) No. of Received Youths

The send-down movement in China

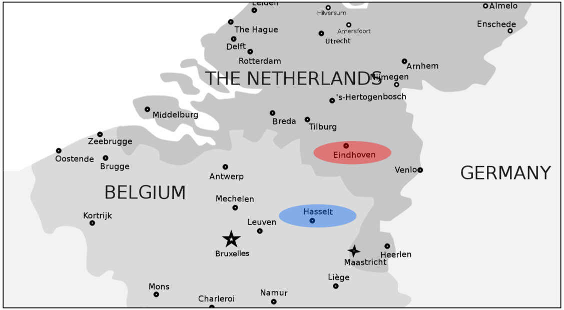
Exposure to sent-down youth has long-lasting effects on educational achievement.



Examples of quasi-experiments...

Bol.com - a Dutch e-restailer - examined the effect of display ads on sales.

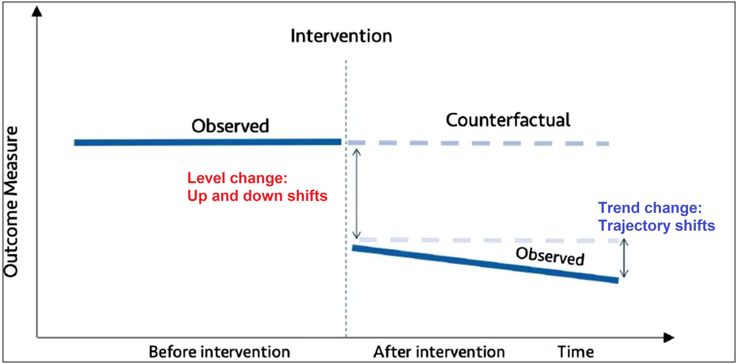
- They compared two cities: one in Holland and one in Belgium.
- They stopped the display ads in the Belgian city.



Before-after design

In practice, we observe an event that happens at a certain time point.

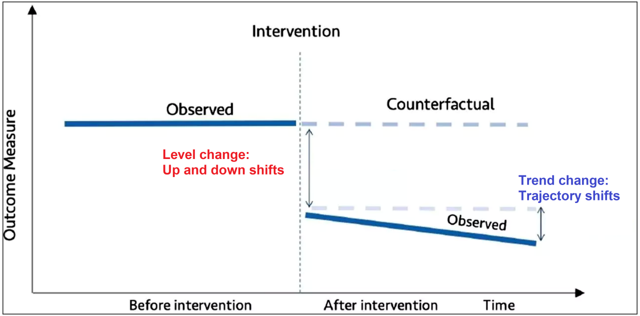
- Predict the "after" outcome based on the before data.
- Compare the "after" outcome with the predicted one.



Before-after design

In practice, **unexpected events occur at random**.

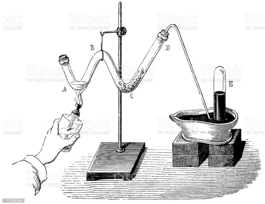
- German reunification
- The implementation of GDPR
- The Great Recession in 2008
- ...



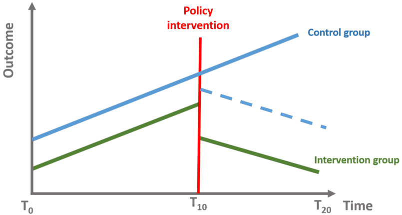
Before-after design

The use of “before” to predict the counterfactual of “after?”

For some scenarios, it’s credible:



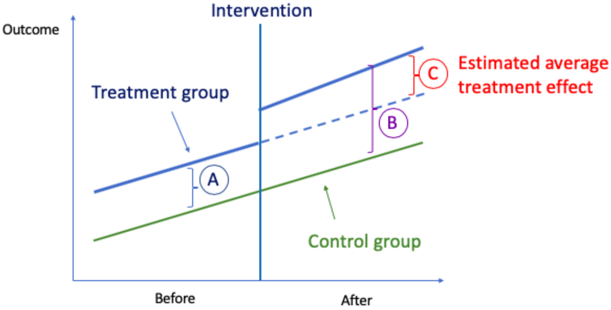
What if the counterfactual also goes through a “regime shift?”



Two sources of variations

Let's add a **control group** for two sources of variation:

- **Time variation**: Before vs. after the intervention.
- **Treatment variation**: Treatment vs. control group.



The canonical DID: basic setup

- N units (i) across $T = 2$ time periods (t).
- A binary treatment $D_{it} = \{0, 1\}$.
- Outcome Y_{it}.

- Treatment happens in period 2 ($t = 2$).
- A group of always untreated units ($D_{it} = 0, \forall t$)

Potential outcomes

2 time periods $\times$ 2 treatment status $=$ 4 potential outcomes.

$Y_{it}(D_i)$	Time 1	Time 2
Untreated	$Y_{i1}(0)$	$Y_{i2}(0)$
Treated	$Y_{i1}(1)$	$Y_{i2}(1)$

The canonical DID: potential outcomes

Observed vs. unobserved potential outcomes.

$Y_{it}(D_i)$	Time 1	Time 2
Untreated	$Y_{i1}(0)$	$\cancel{Y_{i2}(0)}$
Treated	$\cancel{Y_{i1}(1)}$	$Y_{i2}(1)$

(a) Treatment Group

$Y_{it}(D_i)$	Time 1	Time 2
Untreated	$Y_{i1}(0)$	$Y_{i2}(0)$
Treated	$\cancel{Y_{i1}(1)}$	$\cancel{Y_{i2}(1)}$

(b) Control Group

The canonical DID: the target of identification

- **Primary interests**: the treatment effects of the intervention at time period 2.

$$\tau_{\text{ATE}} = E\left(Y_{i2}(1) - Y_{i2}(0)\right)$$

- But, we only observe:

$$\begin{cases} E\left(Y_{i2}(1) \mid D_{i2} = 1\right) & \text{For treatment group} \\ E\left(Y_{i2}(0) \mid D_{i2} = 0\right) & \text{For control group} \end{cases}$$

- Assuming **unconfoundedness**, we can identify τ_{ATE}:

$$\begin{cases} E\left(Y_{i2}(1) \mid D_{i2} = 1\right) & = E\left(Y_{i2}(1)\right) \\ E\left(Y_{i2}(0) \mid D_{i2} = 0\right) & = E\left(Y_{i2}(0)\right) \end{cases}$$

The canonical DID: the target of identification

- The unconfoundedness assumption is unrealistic for quasi-experiments.
- By definition, the treatment is not randomly assigned!

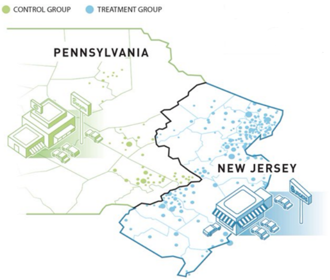
The average treatment effect on the treated (ATT)

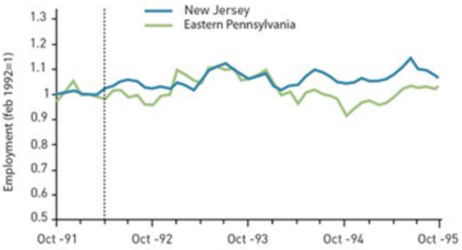
Researchers or policymakers may be more interested in ATT, because:

1. It is uncontrollable who will be in the treatment;
2. Participants of the treatment are more relevant for the policy.

The canonical DID: examples

The influence of minimum wage on employment from Card and Krueger (1994)[1].





[1] Figures are accredited to Jahan Jarnestad at the Royal Swedish Academy of Sciences.

The canonical DID: examples

Snapchat conducted many field experiments on app features and recommendation algorithms.

- They mainly focused on the US market.
- Always used New Zealand as the control group.

The canonical DID: identification assumptions

The target is ATT: $\tau_{\text{ATT}} = E\left[Y_{i2}(1) - Y_{i2}(0) \mid D_{i2} = 1\right]$

- We observe $E\left[Y_{i2}(1) \mid D_{i2} = 1\right]$ for the treatment group.
- But, we do not observe the counterfactual $E\left[Y_{i2}(0) \mid D_{i2} = 1\right]$.

One assumption to make is **ignorability**:

$$E\left[Y_{i2}(0) \mid D_{i2} = 1\right] = E\left[Y_{i2}(0) \mid D_{i2} = 0\right]$$

- The treatment status is independent of $Y_{i2}(0)$.
- Yet, this is still a strong assumption.
- **People often self-select into the control group** (e.g., lack of benefits).

The canonical DID: identification assumptions

It turns out we need **an assumption weaker than ignorability**.

The Parallel Trend Assumption

The before-after difference of the potential outcomes under no treatment is ignorable:

$$\underbrace{E\left[Y_{i2}(0) - Y_{i1}(0) \mid D_{i2} = 1\right]}_{\text{Before-after differences of treatment group}} = \underbrace{E\left[Y_{i2}(0) - Y_{i1}(0) \mid D_{i2} = 0\right]}_{\text{Before-after differences of control group}}$$

Under parallel trend assumption, **ATT of the treatment at $t = 2$ is identified**!

The canonical DID: proof of identification results

Technical proof:
If we rearrange the parallel trend assumption, we have:

$$\begin{aligned} & E\left[Y_{i2}(0) \mid D_{i2}=1\right]-E\left[Y_{i1}(0) \mid D_{i2}=1\right] \\ & -E\left[Y_{i2}(0) \mid D_{i2}=0\right]+E\left[Y_{i1}(0) \mid D_{i2}=0\right]=0 \end{aligned} \tag{1}$$

The ATT is:

$$\tau_{\mathrm{ATT}}=E\left[Y_{i2}(1) \mid D_{i2}=1\right]-E\left[Y_{i2}(0) \mid D_{i2}=1\right]$$

If we add equation (1) to the ATT formula, we have:

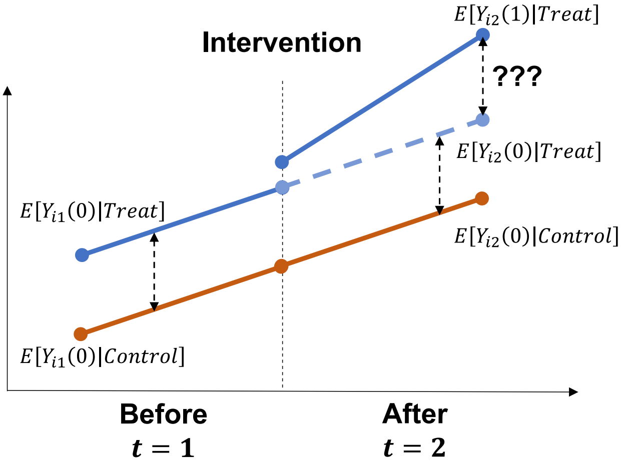
$$\begin{aligned} \tau_{\mathrm{ATT}}= & \left(E\left[Y_{i2}(1) \mid D_{i2}=1\right]-E\left[Y_{i1}(0) \mid D_{i2}=1\right]\right) \\ & -\left(E\left[Y_{i2}(0) \mid D_{i2}=0\right]-E\left[Y_{i1}(0) \mid D_{i2}=0\right]\right) \end{aligned} \tag{2}$$

All the terms in equation (2) can be inferred from the observation.

Understanding the parallel trend

We need to calculate:

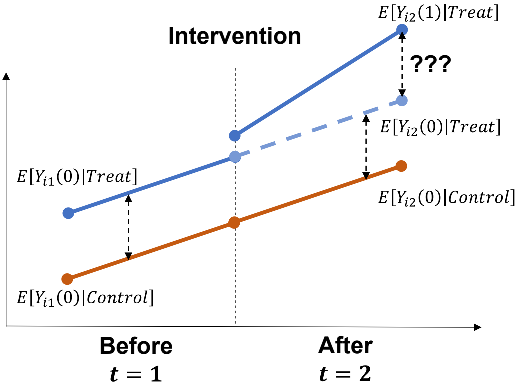
$$\tau_{\mathrm{ATT}} = E\left[Y_{i2}(1) \mid \text{Treat}\right] - \cancel{E\left[Y_{i2}(0) \mid \text{Treat}\right]}$$



Understanding the parallel trend

Counterfactual language: in the absence of treatment, the difference between the treated and control is constant.

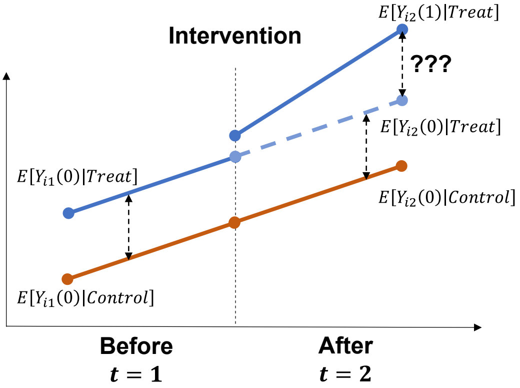
$$E\left[Y_{i1}(0) \mid \text{Treat}\right] - E\left[Y_{i1}(0) \mid \text{Control}\right] =$$
$$\cancel{E\left[Y_{i2}(0) \mid \text{Treat}\right]} - E\left[Y_{i2}(0) \mid \text{Control}\right]$$



Understanding the parallel trend

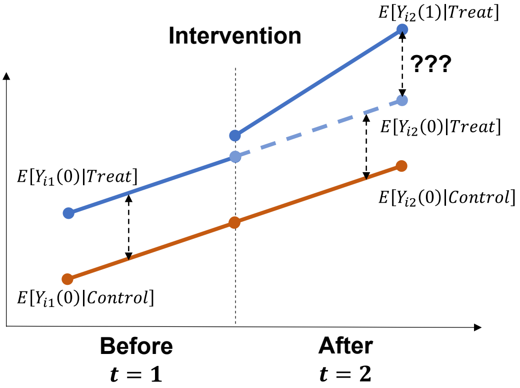
Assume parallel trend to solve for $E\left[Y_{i2}(0) \mid \text{Treat}\right]$.

$$E\left[Y_{i1}(0) \mid \text{Treat}\right] - E\left[Y_{i1}(0) \mid \text{Control}\right] = \underbrace{E\left[Y_{i2}(0) \mid \text{Treat}\right]}_{\text{Solve for this term}} - E\left[Y_{i2}(0) \mid \text{Control}\right]$$



Parallel trend assumption is untestable

- It is an assumption on an unobserved potential outcome $E\left[Y_{i2}(0) \mid \text{Treat}\right]$.
- The fundamental problem of causal inference $\rightarrow$ it's untestable!



Parallel trend assumption is scale-dependent

The parallel trend holds for Y, but may not for a monotonic transformation of Y.

$Y_{it}(D_i)$	Time 1	Time 2
Untreated	$Y_{i1}(0) = 1$	$Y_{i2}(0) = 3$
Treated	$Y_{i1}(0) = 2$	$Y_{i2}(0) = 4$

(a) Original Values: $2 - 1 = 4 - 3$

$Y_{it}(D_i)$	Time 1	Time 2
Untreated	$log(Y_{i1}(0)) = log(1)$	$log(Y_{i2}(0)) = log(3)$
Treated	$log(Y_{i1}(0)) = log(2)$	$log(Y_{i2}(0)) = log(4)$

(b) Log-transformed Values: $log(2) \neq log\left(\frac{4}{3}\right)$

DID as an experimental design

DID does *not* require random assignment of treatment to units.

- The target estimand is usually the **ATT**.
- No need for the treated and control group to be exchangeble.
- Parallel trend $\rightarrow$ a weaker design condition about **untreated trends**.

Key point

The control group does not need to be a randomized copy of the treated group. It needs to provide the right counterfactual *trajectory*.

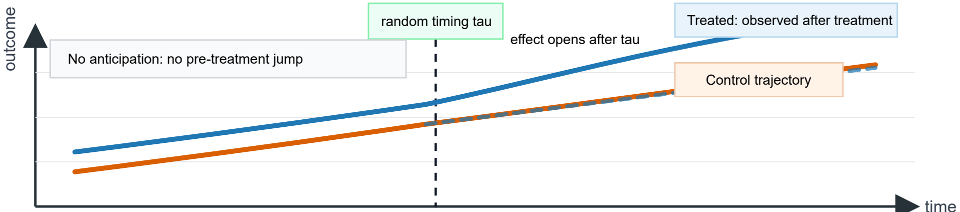
Experimental design that makes parallel trends credible

The best design is often not randomizing *who* is treated, but making the *timing* of treatment as good as random.

Ideal DID design

Treatment starts at a time that is unrelated to the untreated outcome trajectory.

$$P(\mathbf{D}_{it} \mid G_i, \Delta Y_i(0)) = P(\mathbf{D}_{it} \mid G_i)$$



Non-parametric estimation

From the identification results (see slide 27):

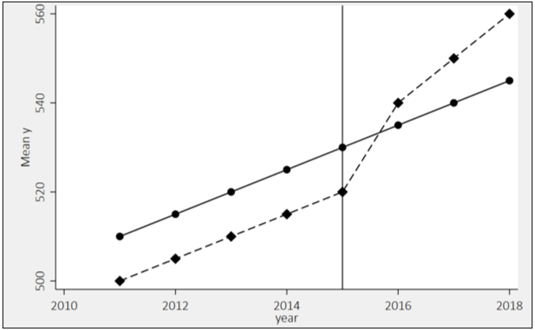
$$\tau_{\text{ATT}} = \left(E\left[Y_{i2}(1) \mid \text{Treat}\right] - E\left[Y_{i1}(0) \mid \text{Treat}\right]\right) \\ - \left(E\left[Y_{i2}(0) \mid \text{Control}\right] - E\left[Y_{i1}(0) \mid \text{Control}\right]\right),$$

A natural non-parametric estimator is

$$\hat{\tau}_{\text{ATT}} = \left(\frac{1}{N_1}\sum_{i\in\text{Treat}} Y_{i2} - \frac{1}{N_1}\sum_{i\in\text{Treat}} Y_{i1}\right), \\ - \left(\frac{1}{N_0}\sum_{i\in\text{Control}} Y_{i2} - \frac{1}{N_0}\sum_{i\in\text{Control}} Y_{i1}\right)$$

Two differences $\Rightarrow$ “difference-in-differences”

Multiple time periods



Average the time periods and get 4 conditional expectations.

- Not efficient as aggregation loses information.

Model-based estimation: two-way fixed effects regression.

Model-based estimation

A two-way fixed effects regression:

$$y_{it} = \alpha_i + \lambda_t + \underbrace{\tau}_{\text{ATT}}(\text{Treated}_i \times \text{After}_t) + \varepsilon_{it}$$

- α_i: individual fixed effects.
- λ_t: time fixed effects.
- After_t: a time dummy of before vs. after treatment.
- Treated_i: an individual dummy of treated vs. control.

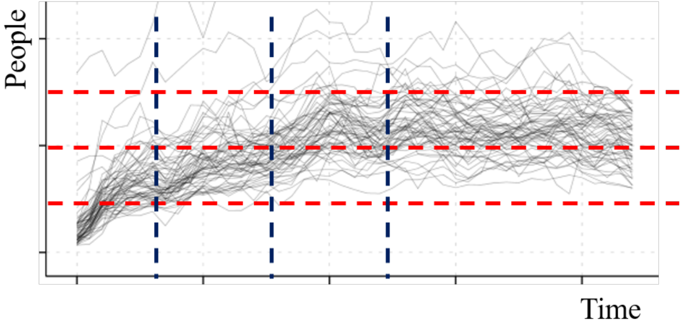
Model-based estimation

To better understand the model:

$$y_{it} = \alpha_i + \lambda_t + \tau\left(\text{Treated}_i \times \text{After}_t\right) + \varepsilon_{it}$$

	Before ($t = 1$)	After ($t = 2$)	Difference
Control ($i = 1$)	$\alpha_1 + \lambda_1 + \varepsilon_{11}$	$\alpha_1 + \lambda_2 + \varepsilon_{12}$	$E\left(\Delta Y_C\right) = \lambda_2 - \lambda_1$
Treated ($i = 2$)	$\alpha_2 + \lambda_1 + \varepsilon_{21}$	$\alpha_2 + \lambda_2 + \tau + \varepsilon_{22}$	$E\left(\Delta Y_T\right) = \tau + \lambda_2 - \lambda_1$
DID			$E\left(\Delta Y_T\right) - E\left(\Delta Y_C\right) = \tau$

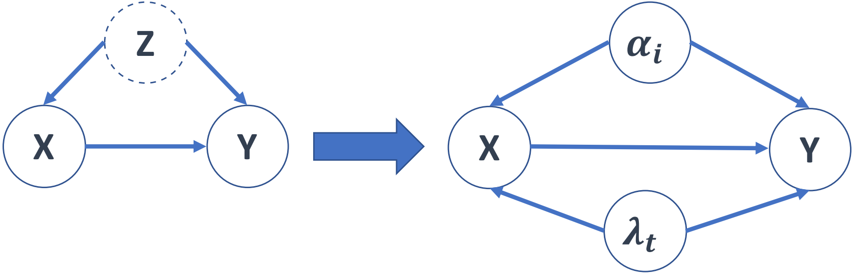
Model-based estimation: a link to data



With the two dimensional data (time-person), we can "control for"

1. Individual fixed effects (α_i) from repeated time periods;
2. Time fixed effects (λ_t) from multiple people.

Another look at the identification of DID



- DID approach is by no means fail-safe!
- Key threats to identification: **confounders that vary across individuals and time**.

How to examine the parallel trend assumption?

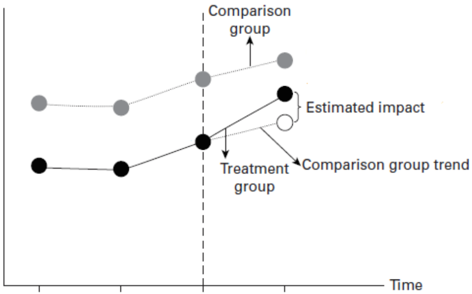
The parallel trend assumption is untestable! What to do then?

$$\underbrace{E\left[Y_{i1}(0) \mid \text{Treat}\right] - E\left[Y_{i1}(0) \mid \text{Control}\right]}_{\text{Pre-treatment trend}} = \underbrace{\color{red}{E\left[Y_{i2}(0) \mid \text{Treat}\right]} - E\left[Y_{i2}(0) \mid \text{Control}\right]}_{\text{Post-treatment trend}}$$

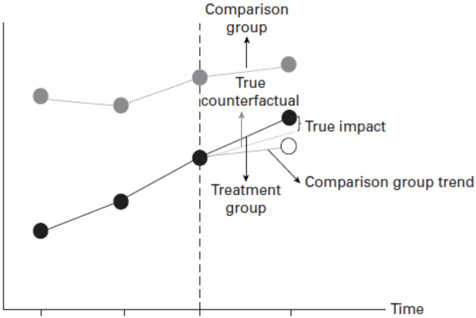
- We cannot observe the post-treatment trend, due to the lack of $\color{red}{E\left[Y_{i2}(0) \mid \text{Treat}\right]}$.
- **But** we do observe the pre-treatment trend.
- **Can we somehow use the pre-trends?**

How to examine the parallel trend assumption?

Use your intuitions to compare two DID analysis:



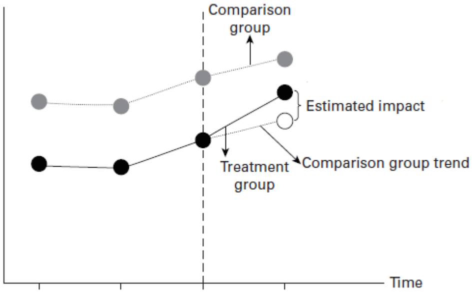
(a) Parallel Pre-trends



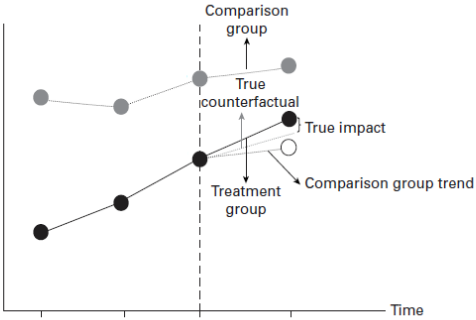
(b) Non-parallel Pre-trends

How to examine the parallel trend assumption?

Intuition: **Observing similar pre-trends may increase our confidence that the parallel assumption holds**.



(a) Parallel Pre-trends



(b) Non-parallel Pre-trends

Discussions on the intuition of parallel pre-trends

- Is the parallel pre-trend a sufficient condition for the parallel trend assumption?

Discussions on the intuition of parallel pre-trends

- Is the parallel pre-trend a sufficient condition for the parallel trend assumption?
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Discussions on the intuition of parallel pre-trends

- Is the parallel pre-trend a sufficient condition for the parallel trend assumption?
- Is the parallel pre-trend a necessary condition for the parallel trend assumption?
- Does the pre-trend test increase the plausibility of the parallel trend assumption?

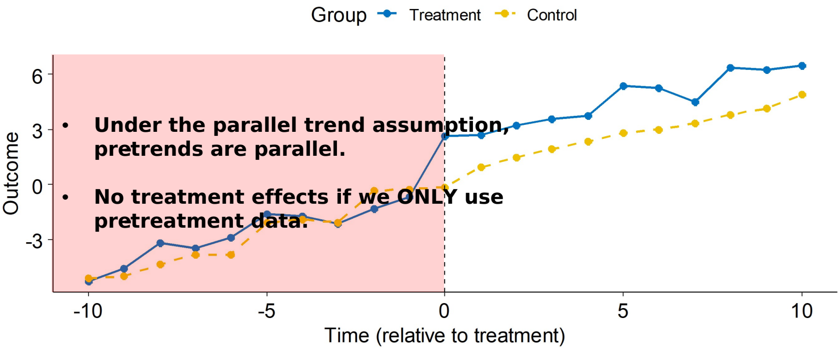
Discussions on the intuition of parallel pre-trends

- Is the parallel pre-trend a sufficient condition for the parallel trend assumption?
- Is the parallel pre-trend a necessary condition for the parallel trend assumption?
- Does the pre-trend test increase the plausibility of the parallel trend assumption?
- Think about the difference between a t-test of means, an ANOVA of the treatment effect, and the pre-trend test.

Pre-trend testing with placebo tests

Definition (Placebo Tests)

A pseudo-treatment (like a placebo in a drug trial) should not or cannot have an effect. Therefore, analyses with the pseudo-treatment should yield null results.



Placebo tests in DID

The procedure of placebo tests in DID

1. Obtain the pre-treatment data.
2. Construct a pseudo treatment $\widetilde{\text{After}}_t$ by assuming the treatment happens at t in the pre-data.
3. Run a two-way fixed effects regression with $\widetilde{\text{After}}_t$ on pre-data.

$$y_{it} = \alpha_i + \lambda_t + \tilde{\tau}\left(\text{Treated}_i \times \widetilde{\text{After}}_t\right) + \varepsilon_{it}$$

4. Repeat for different times t and collect the pseudo-ATT $\tilde{\tau}$.

Under the parallel trend, $\tilde{\tau}$ should be statistically 0.

Illustrating trends with event study graphs

Another way is to look at the dynamics of treatment effects using **event study graphs**.

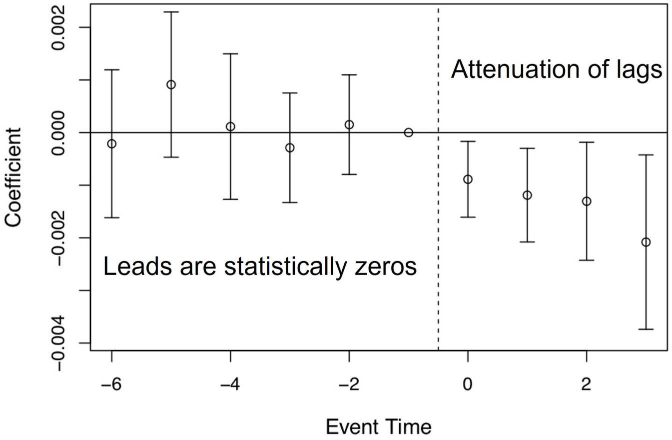
Event Study Graphs

- Estimate a two-way fixed effects regression with q leads, m lags and the original treatment timing After_t.

$$y_{it} = \alpha_i + \lambda_t + \sum_{s=-q}^{m} \widetilde{\tau}_s \left(\text{Treated}_i \times \left(\widetilde{\underbrace{L^{-s}}_{\text{Lag}} \text{After}_t} \right) \right) + \varepsilon_{it}$$

- Plot the coefficients $\widetilde{\tau}_s$ and check the patterns.

Illustrating trends with event study graphs



Sensitivity analysis with pre-trends

Decomposing the estimated ATT:

$$
\begin{aligned}
\hat{\tau}_{\mathbf{ATT}} &= E\left(Y_{i2}^{1} - Y_{i1}^{0} \mid D_i = 1\right) - E\left(Y_{i2}^{0} - Y_{i1}^{0} \mid D_i = 0\right) \\
&= \underbrace{E\left(Y_{i2}^{1} - Y_{i2}^{0} \mid D_i = 1\right)}_{\textbf{True } \tau_{\mathbf{ATT}}} + \\
&\quad \underbrace{\left[E\left(Y_{i2}^{0} - Y_{i1}^{0} \mid D_i = 1\right) - E\left(Y_{i2}^{0} - Y_{i1}^{0} \mid D_i = 0\right)\right]}_{\textbf{Difference in Trends}}
\end{aligned}
$$

From this, we have the estimation bias is:

$$
\hat{\tau}_{\mathrm{ATT}} - \tau_{\mathrm{ATT}} = \underbrace{E\left(Y_{i2}^{0} - Y_{i1}^{0} \mid D_i = 1\right) - E\left(Y_{i2}^{0} - Y_{i1}^{0} \mid D_i = 0\right)}_{\text{Difference in Trends}}
$$

Sensitivity analysis with pre-trends

$$\hat{\tau}_{\text{ATT}} - \tau_{\text{ATT}} = \underbrace{E\left(Y_{i2}^0 - Y_{i1}^0 \mid D_i = 1\right) - E\left(Y_{i2}^0 - Y_{i1}^0 \mid D_i = 0\right) \equiv \delta_1}_{\text{Difference in Trends}}$$

- Under parallel trend or $\delta_1 = 0$, we have no bias.

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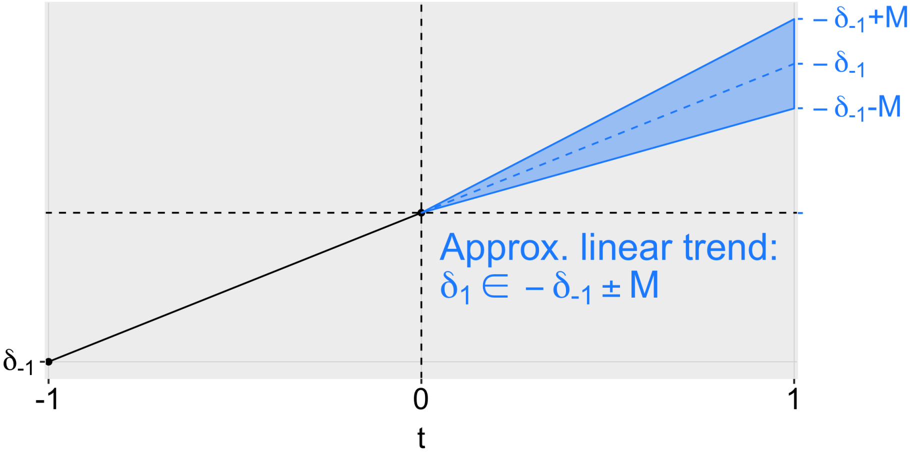
- Under parallel trend or $\delta_1 = 0$, we have no bias.

- Now assume the parallel trend is violated or $\delta_1 \neq 0$.
- **Q: what extent of the violation (δ_1) would nullify the τ_{ATT}?**
  - A larger value of δ_1 implies a more credible $\hat{\tau}_{\text{ATT}}$.

Sensitivity analysis with pre-trends

Illustrating the idea of sensitivity analysis:



Sensitivity analysis with pre-trends

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Problem: how to set the values of δ_1?

It seems fairly random to find proper values of $\delta_1 \in (-\infty, +\infty)$.

How to find a good anchor for the values of δ_1?

Sensitivity analysis with pre-trends

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How to find a good anchor for the values of δ_1?

Solution: using pre-trends as anchors

Assuming δ_1 is X times the values of the difference in pre-trends.
X is interpretable and applicable to different studies.

Sensitivity analysis with pre-trends

Consider a DID with 3 time periods with $t = 0, 1, 2$, and the treatment occurs in $t = 2$.

Sensitivity analysis with pre-trends

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1. The difference in pre-trends δ_0 is:

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$$|\delta_1| < M\,|\delta_0|$$

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2. Assume the difference in post-trends δ_1 is bounded as:

$$|\delta_1| < M\,|\delta_0|$$

3. We can vary the value of M and get a critical value M^* that nullifies the ATT.

What if pre-trends are not parallel?

You need extra assumptions / information for a more credible identification.

- **Scenario 1**: With pre-treatment characteristics X_i.
- **Scenario 2**: With outcomes Y_{it} of many pre-treat periods.

Parallel pre-trends

Parallel pre-trends are **neither necessary or sufficient** for the parallel trend assumption!

Scenario 1: With pre-treatment variables X_i

Conditional Parallel Trends Assumption

Conditioning on X_i, we have the following:

$$E\left[Y_{i2}(0) - Y_{i1}(0) \mid D_i = 1, X_i\right] = E\left[Y_{i2}(0) - Y_{i1}(0) \mid D_i = 0, X_i\right]$$

Scenario 1: With pre-treatment variables X_i

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Identification results

Given the assumption, we can identify and conditional ATT:

$$\tau_{\text{ATT}}(x) = \underbrace{E\left[Y_{i2} - Y_{i1} \mid D_i = 1, X_i = x\right]}_{\text{Changes for the treated at } x} - \underbrace{E\left[Y_{i2} - Y_{i1} \mid D_i = 0, X_i = x\right]}_{\text{Changes for the control at } x}$$

Scenario 1: With pre-treatment variables X_i

Matching (see Heckman et al. 1997):

$$\hat{\tau}_{\text{ATT}} = \frac{1}{N_1} \sum_{i \in \text{Treated}} \Big((Y_{i2} - Y_{i1}) - \hat{E}_X \left[Y_{i2} - Y_{i1} \mid D_i = 0, X_i \right] \Big)$$

- $\hat{E}_X \left[Y_{i2} - Y_{i1} \mid D_i = 0, X_i \right]$ is the estimated expectation function fitted on the control units.
- It takes in values of X_i and predicts the counterfactual changes at X_i for a treated unit.
- In theory, any function form can be used, e.g., a polynomial function.

Scenario 1: With pre-treatment variables X_i

Weighting (see Abadie 2005):

$$\hat{\tau}_{\text{ATT}} = \frac{1}{N_1} \sum_{i \in \text{Treated}} (Y_{i2} - Y_{i1}) - \frac{1}{N_0} \sum_{i \in \text{Control}} \left(\frac{\hat{e}(X_i)}{1 - \hat{e}(X_i)} \right) (Y_{i2} - Y_{i1})$$

- $\hat{e}(X_i)$ is the estimated propensity score function.
- We first estimate $\hat{e}(X_i)$ and then plug it in.

Scenario 1: With pre-treatment variables X_i

With multiple time periods, we can run a two-way fixed effects regression with X_i as controls:

$$y_{it} = \alpha_i + \lambda_t + \tau\,(\text{Treated}_i \cdot \text{After}_t) + \beta\,(X_i \cdot \text{After}_t) + \varepsilon_{it}$$

- Note that this specification may not be consistent – no treatment heterogeneity.
- The coefficient β is the same across different levels of X_i.
- With too many X_i, we can use sub-classification based on X_i.

Scenario 2: With many pre-treatment periods

In weighting approach, we weight the control units for ATT (ω^{SDID}).

- Inadequate if the pre-trend differ in some periods but not all.

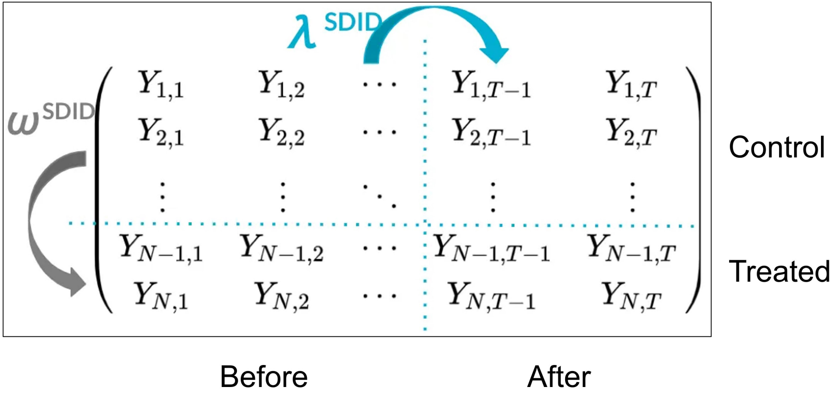
$$\omega^{\text{SDID}} \begin{pmatrix} Y_{1,1} & Y_{1,2} & \cdots & Y_{1,T-1} & Y_{1,T} \\ Y_{2,1} & Y_{2,2} & \cdots & Y_{2,T-1} & Y_{2,T} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ Y_{N-1,1} & Y_{N-1,2} & \cdots & Y_{N-1,T-1} & Y_{N-1,T} \\ Y_{N,1} & Y_{N,2} & \cdots & Y_{N,T-1} & Y_{N,T} \end{pmatrix} \begin{array}{l} \\ \text{Control} \\ \\ \\ \text{Treated} \end{array}$$

Before After

Scenario 2: With many pre-treatment periods

Arkhangelsky 2021 proposes **synthetic DID** to also weight the time periods (λ^{SDID}).

- Calculate λ^{SDID} with long pre-treatment periods.
- Integrating the idea of synthetic control and DID analysis.

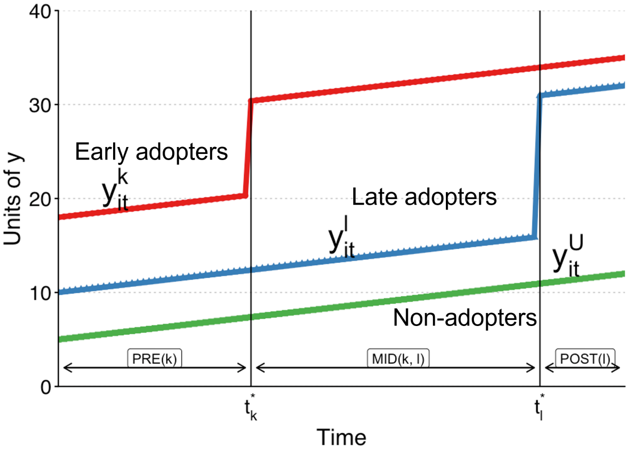


Differential timing of the treatment

So far, we assume two things about the treatment:

1. Treatment happens at t for all people.
2. Once treated, remain treated[a].

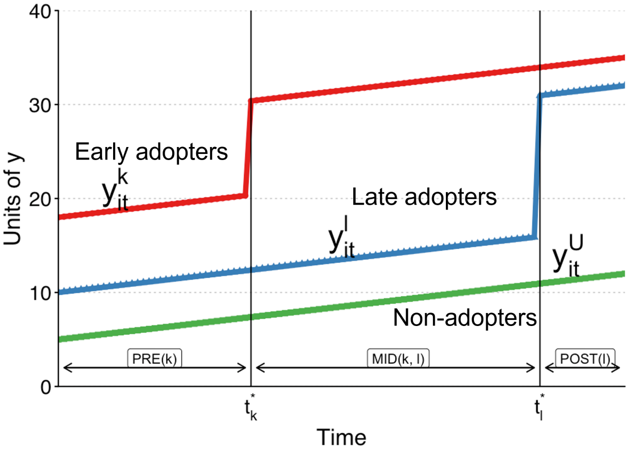
[a] In statistical term, treatment is an absorbing state.



Differential timing of the treatment

In practice, we have **"staggered adoptions"**: groups receive treatments at different times.

- Product launches
- Stock buybacks and dividend policies
- Minimum wage laws
- Vaccination programs
- . . .

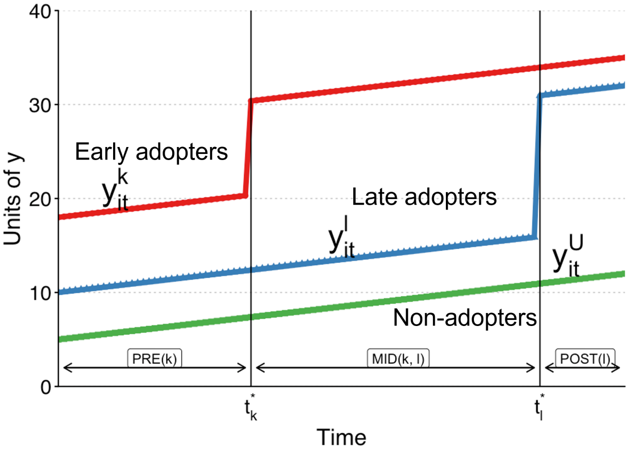


Differential timing of the treatment

Can we still use the two-way fixed effects regression?

$$y_{it} = \alpha_i + \lambda_t + \tau \underbrace{D_{it}}_{\text{Treatment}} + \varepsilon_{it}$$

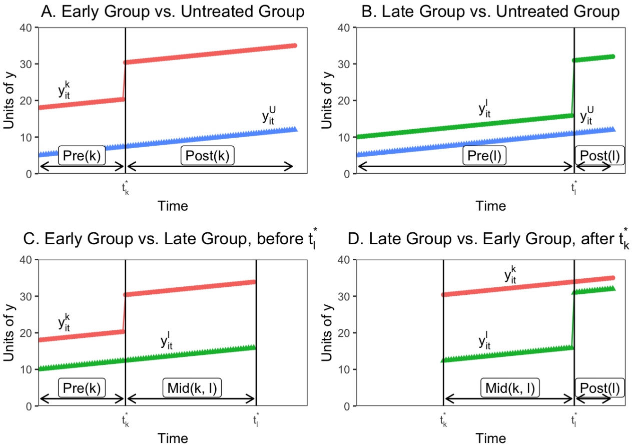
- Instead of $(\text{Treated}_i \times \text{After}_t)$, we have a variable D_{it}.
- $D_{it} \in \{0, 1\}$ combines group-on-group comparisons.



Differential timing of the treatment

Goodman-Bacon (2021) formalizes $D_{it} \in \{0, 1\}$ as group-on-group comparisons.

- Not all comparisons are valid.
- The existence of **forbidden comparisons**.



How to correct for the bias?

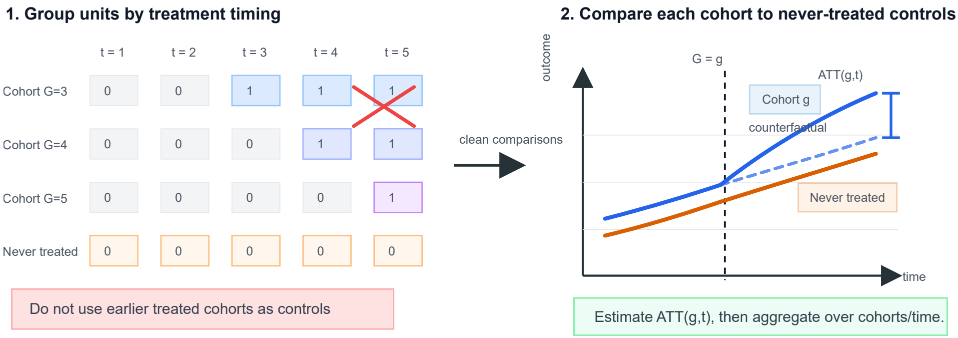
A few approaches proposed recently:

- Callaway, B., & Sant'Anna, P. H. (2021). difference-in-differencess with multiple time periods. Journal of Econometrics, 225(2), 200-230.
- Sun, L., & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. Journal of Econometrics, 225(2), 175-199.
- Cengiz, D., Dube, A., Lindner, A., & Zipperer, B. (2019). The effect of minimum wages on low-wage jobs. The Quarterly Journal of Economics, 134(3), 1405-1454.

Staggered DID: clean comparisons

To avoid forbidden comparisons:

1. Group treated units by the same treatment timing.
2. For each cohort g, compare them only with never-exposed units.



Staggered DID: assumptions

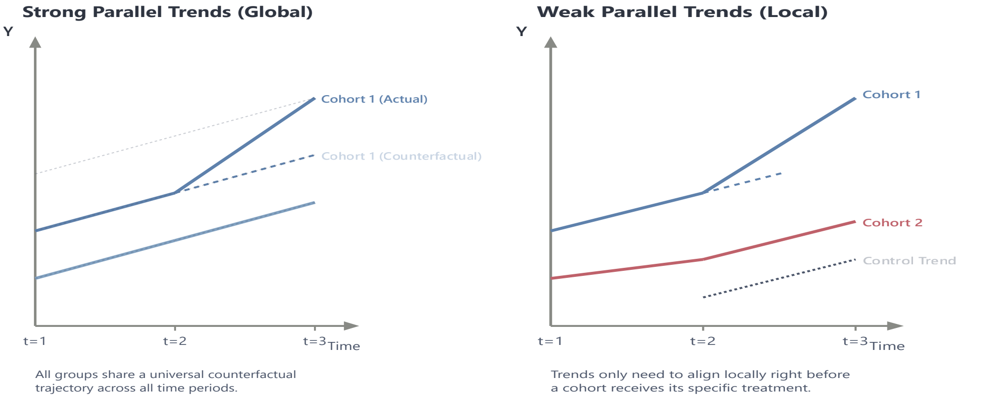


Figure 7: Strong vs. weak parallel trend assumptions

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